A Paleomagnetic Reversal in the Osler Volcanic Group, Northern Lake Superior

H. C. HALLS
Department of Geology, Erindale College, University of Toronto

Received February 5, 1974
Revision accepted for publication May 6, 1974

A geological and paleomagnetic study in the Nipigon Strait area, northern Lake Superior, has confirmed previous conclusions drawn from aeromagnetic data, that an unconformity occurs within the Late Precambrian Keweenawan Osler volcanics and separates normally magnetized lavas from older ones with reversed polarity. The new data, together with other paleomagnetic and geological evidence from Keweenawan rocks, suggest that the magnetic reversal occurred when there was a temporary halt or decline in volcanic activity throughout the Lake Superior region. During the quiescent period, coarse clastic sediments were deposited by marginal erosion of the subsiding Keweenawan basin; sinking of the basin with corresponding uplift of the margins may have been greater in the south where the thickest sections of conglomerate and sandstone are preserved.

Une étude géologique et paléomagnétique dans la région du Détroit de Nipigon dans la partie Nord du Lac Supérieur a confirmé des conclusions antérieure tirées de données aéromagnétiques à savoir qu'une discordance existe à l'intérieur des roches volcanique Keweenawan Osler du précambrien tardif et sépare des laves normalement magnétisées d'autres laves plus vieilles à magnétisme inversé. Les nouvelles données ainsi que celles antérieures en plus des évidences géologiques concernant les roches Keweenawan suggèrent que l'inversion magnétique s'est produite lors d'une halte temporaire ou lors d'un déclin dans l'activité volcanique de toute la région du Lac Supérieur. Pendant les périodes de tranquilité des sédiments clastiques grossiers furent déposés par érosion marginale du bassin Keweenawan en subsidence; l'enfoncement du bassin ainsi que l'élévation de la bordure peuvent avoir été plus marqués dans le Sud là où sont préservées les sections les plus épaisses de conglomérat et de grès. [Traduit par le journal]

Introduction

An analysis of aeromagnetic data from the north shore region of Lake Superior suggests (Halls 1972) that an unconformity separating lavas of normal and reversed magnetic polarity occurs near the top of the Late Precambrian Osler Volcanic Group that forms most of the mainland and outer islands east of Thunder Bay (Fig. 1). The Osler Group has a maximum exposed thickness of about 6000 ft (1800 m) and dips lakeward at angles ranging from less than 5° to over 30° (McIlwaine *et al.* 1971).

To test the validity of the aeromagnetic interpretation a geological and paleomagnetic study was made of the volcanics at the mouth of Nipigon Strait (Fig. 1) where the youngest rocks of the Osler Group are exposed. This paper presents the results of the investigation.

Geology of the Nipigon Strait Area

Stratigraphy

The Osler volcanics form the lowest part of the extensive Keweenawan series which borders and underlies most of Lake Superior (Fig. 1). The dominant units in the Keweenawan are a thick sequence of tholeiitic flood basalts (locally in excess of 20 000 ft (6000 m)) overlain by a comparably thick clastic sedimentary succession. The rocks generally dip lakeward so that they form a large, irregularly-shaped basin (White 1966; Halls 1966; Halls and West 1971) which lies at the apex of a long (~ 1000 miles (~ 1600 km)) horseshoe-shaped belt of Keweenawan rocks, the arms of which extend southward into the central U.S.A. beneath a Paleozoic cover. The origin of Keweenawan rocks has been generally ascribed to a rifting mechanism similar to that which has led to continental fragmentation in more recent times (*e.g.*, Smith *et al.* 1966; White 1972; Chase and Gilmer 1973).

The uppermost 4000 ft (1220 m) of the Osler Group, exposed at the entrance to Nipigon Strait (Figs. 2 and 3) are composed mainly of grey-black massive basalt flows with reddish amygdular tops, together with a coarse-grained diabase and red or grey feldspar por-

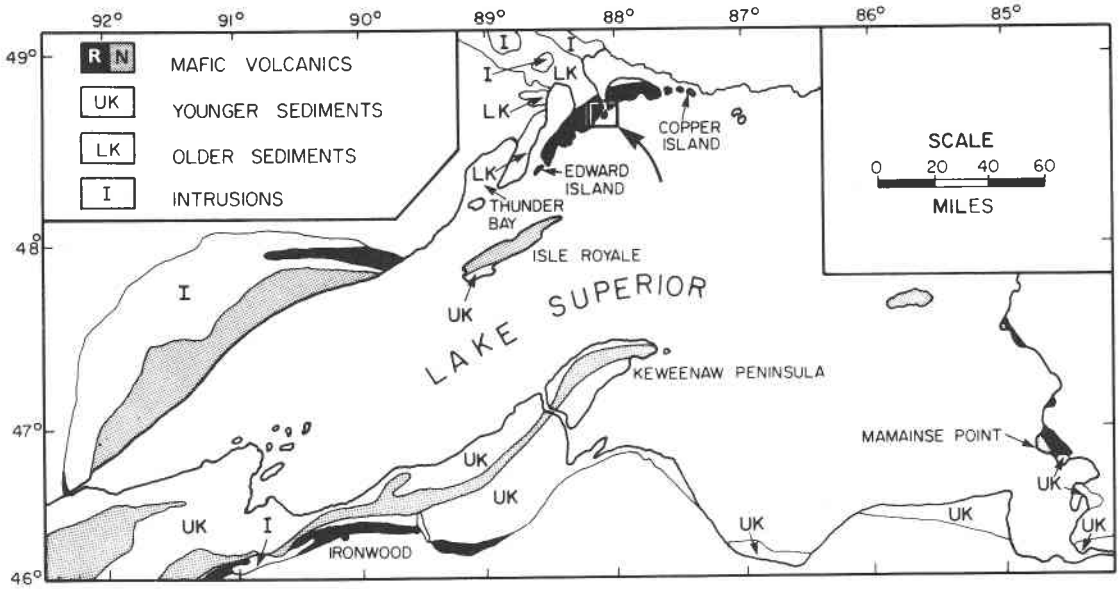


FIG. 1. Map showing distribution of Keweenawan rocks around Lake Superior and location of Nipigon Strait survey area (arrowed). R = Reversely magnetized; N = Normally magnetized. Magnetic data are summarised from Palmer (1970), Books (1968, 1972), and Books and Green (1972). A thin, normally magnetized sequence of volcanics and clastics occurring at the base of the reversely magnetized lavas in Michigan and Minnesota (Books 1972; Books and Green 1972) is not shown.

phyry which form about a third of the sequence. The porphyry appears to be extrusive near Spar Island (Fig. 2) but its exact nature in the vicinity of Fluor Island is unclear. The diabase forming the southwestern end of Fluor Island is likewise of unknown nature but it may be part of the Moss Lake gabbro intrusion to the northwest (McIlwaine *et al.* 1971).

The mafic flows range in thickness from a few feet to more than a hundred. The thickest flow encountered is a coarse-grained purple-brown basalt which occurs on Puff Island (Fig. 3). It rests on a 30-ft (9 m) thick conglomerate carrying pebbles of porphyry and basalt, and is overlain by a series of thin, red, flaggy basalt flows which form the southwest shore of the island. The lava sequence extends along strike to the southwest; red flaggy basalts outcrop on Tremblay Island and massive basalt identical in appearance to the thick flow on Puff Island occurs on Cedar and Provost Islands (Fig. 3). The conglomerate pinches out westwards as it is not seen at the southwest end of Puff Island.

Structure

With the exception of the conglomerate and a sediment-filled channel cut in a flow top exposed on the northwest shore of Puff Island, no other interflow sediments were observed. The scarcity of these rocks within the volcanic sequence hinders reliable estimation of structural dip necessary for paleomagnetic data. Fortunately most flows show a crude but pronounced columnar jointing which appears perpendicular to the flow base where ever the two features are visible. The attitude of columnar jointing was thus used to assess dip at most paleomagnetic sites.

Throughout the area shown in Fig. 2 the volcanics are tilted about 30°SE but have substantially lower dips of about 10°SE above the conglomerate and below the diabase–porphyry zone. Figure 3 shows an angular discordance in dip of about 20° between flows above and below the Puff Island conglomerate. Dip determinations were made at the junction of the conglomerate with the overlying flow and from columnar jointing in a flow about 600 ft (180 m) north of the contact and hence below the conglomerate. Gradational dips may occur in the intervening region of poor exposure, but the presence of the conglomerate suggests that the apparent discordance is real and that an

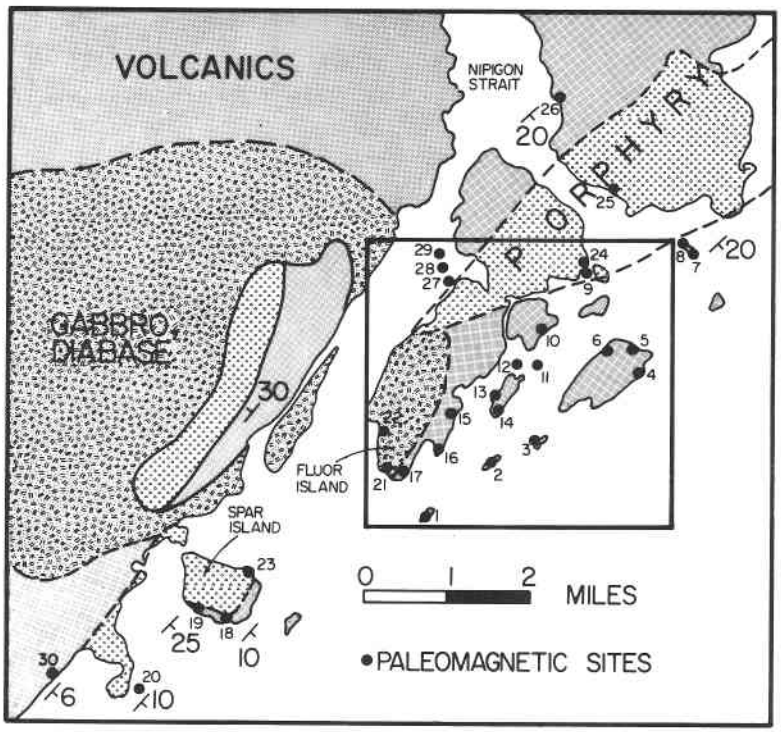


FIG. 2. Geological map of the Nipigon Strait area, showing location of paleomagnetic sites. Area within square outline is shown enlarged in Fig. 3. Geological data modified after McIlwaine *et al.* (1971).

unconformity exists. Further support for an unconformity is indicated in Fig. 3 by a pronounced southwesterly increase in angle between the strike directions of flows stratigraphically above and below the conglomerate; flows at the southern end of Fluor Island strike 20°, whereas rocks on Cedar and Provost Islands immediately to the south strike 50°. Columnar jointing on these outermost islands suggests a lakeward dip of only about 10°, considerably less than on Fluor Island.

Geological data thus support conclusions previously reached from aeromagnetic data (Halls 1972) that an unconformity exists within the Osler Group in the Edward and Copper Island areas (Fig. 1). The new observations suggest that the discordant contact probably extends along the entire length of the Lake Superior shoreline between the two islands.

Paleomagnetic Investigations

Method and Results

In the present investigation a total of 30 paleomagnetic sites were obtained (Fig. 2), each comprising three to six independently oriented samples. For seven sites the samples were blocks, oriented using a magnetic compass, while for all remaining sites core samples were obtained using a portable field drill, drilling platform, and sun compass.

The platform was a rectangular aluminum plate with adjustable screw legs. In the center of the plate and perpendicular to it was a circular collar about 8 in. (20 cm) high, lined with a lubricated, plastic-based sleeve bearing (type DX28D from Garlock Bearings Inc.) which was capable of withstanding frictional temperatures in excess of 400°C. The shaft of the drill (a modified 2 H.P. McCulloch chain saw) was lengthened with an adaptor which could pass through the collar with a tolerance of only 0.001 in. (0.00254 cm). A flat rock surface was selected and a north arrow marked on it (to within ± 2° of azimuth) using the sun compass. The platform was then centered over the desired drilling spot, levelled, and coring completed by the operator standing on the platform to stabilize it. With proper platform levelling, the core

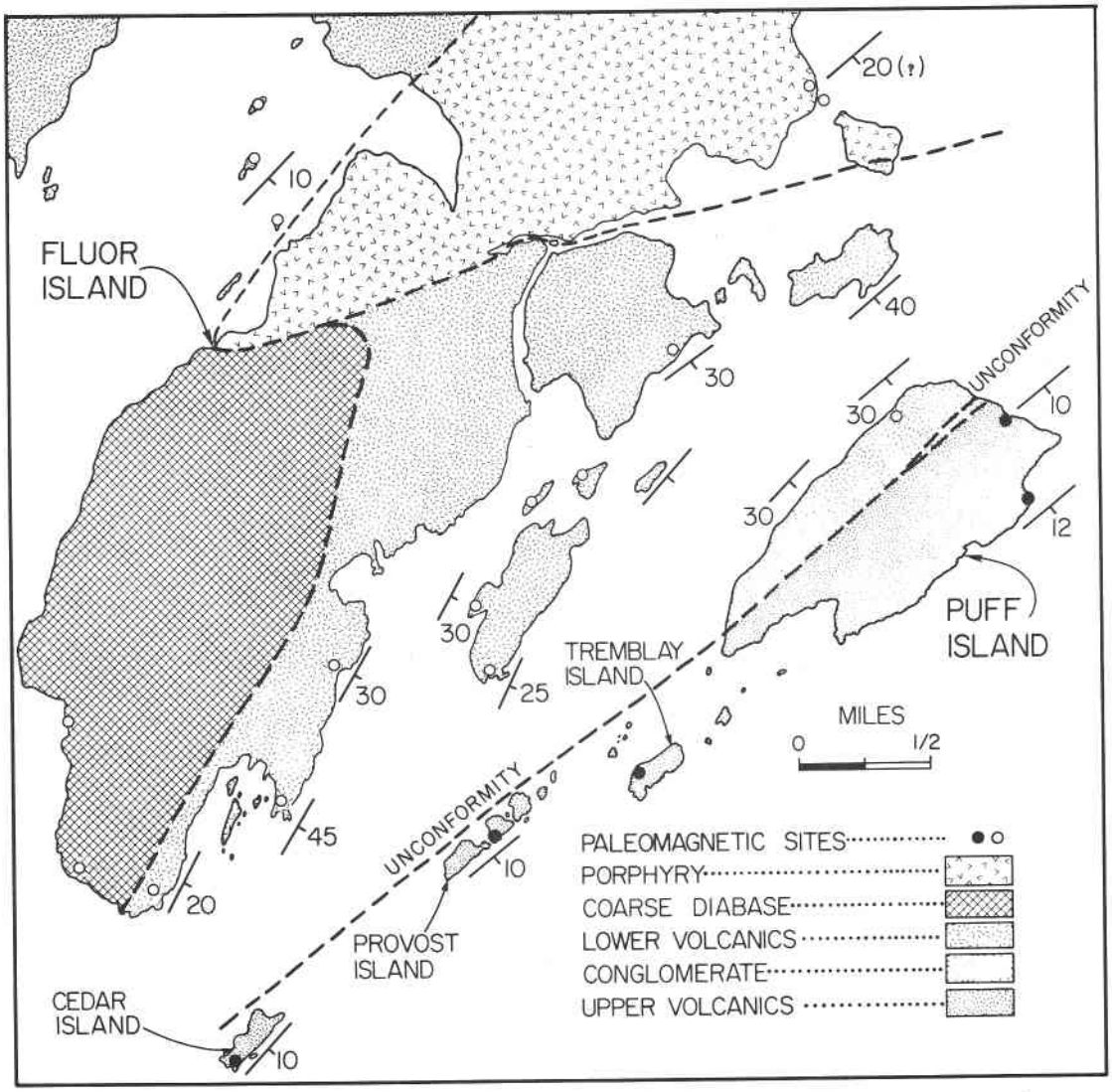


FIG. 3. Detailed geological map of the Fluor Island area, showing position of suspected unconformity. Paleomagnetic sites of normal and reversed polarity are shown by solid and open circles respectively.

barrel was constrained to move through the sleeve bearing to within 1° of the vertical. The advantage of the above method is that smooth, straight, vertical cores can be obtained with no loss should the core break during retrieval of the drill barrel.

For each sample, at least three 1 in. × 1 in. (2.5 cm × 2.5 cm) core specimens were cut and their NRM intensities and directions measured using a PAR spinner magnetometer. At least one specimen per sample was progressively demagnetized to maximum a.c. fields of 400 oe, using 50 oe field increments between 100 and 300 oe. The optimum cleaning field for each site was that which gave the highest precision estimate k (Fisher 1953) for sample magnetization directions.

A complete list of paleomagnetic results at the site and mean site levels is given in Tables 1 and 2 respectively, and equal area projections summarising the paleomagnetic directions are shown in Fig. 4.

Interpretation of Results

On demagnetization, site directions tend to swing to the west or northwest (Fig. 4A), but

TABLE 1. Summary of site data for Osler volcanics in the Nipigon Strait area
(Those sites with an * were obtained using a magnetic compass)

Site	N	$D_1°$	$I_1°$	H	k	δ°	ω°	$D_2°$	$I_2°$	∂°	J_0	J_{200}/J_0 (%)	J_{400}/J_{200} (%)
1	5	293.1	34.5	200	517	45	10	289.8	43.6	8	1.95	67.2	54.2
2	5	290.6	31.9	150	243	50	12	285.7	42.0	10	3.57	38.9	28.1
3	5	310.0	24.6	100	2485	56	14	307.4	37.9	7	2.22	92.4	83.2
4	5	303.9	18.2	150	331	45	12	302.8	29.9	10	1.11	92.4	82.0
5	5	297.3	33.1	300	130	45	10	294.7	42.5	4	3.33	40.7	36.3
6	5	112.6	−18.1	100	70	41	28	106.1	−44.2	14	2.01	11.4	38.5
7	6	124.5	−45.3	250	351	59	20	109.8	−62.4	6	4.08	87.9	73.4
8	5	144.3	−47.0	400	679	60	18	140.7	−64.8	15	12.28	41.6	40.3
9	3	121.0	−39.1	150	321	55	20	110.0	−56.6	3	4.12	55.5	32.3
10	5	129.7	−12.1	150	59	44	30	128.2	−42.0	17	3.75	52.7	40.0
11	3	106.7	−33.9	400	456	42	30	88.3	−59.1	14	11.86	27.9	19.8
12	5	123.2	−09.6	250	240	42	40	118.6	−48.9	8	5.32	47.0	47.2
13	3	126.4	−03.3	250	886	30	35	128.1	−38.0	21	4.01	42.4	43.1
14	6	111.9	−36.2	200	121	30	20	108.2	−55.8	4	1.31	28.2	11.2
15	6	117.6	−16.1	300	402	25	30	118.5	−46.0	11	1.87	40.5	31.4
16	6	107.7	−09.3	150	125	20	45	106.0	−54.2	6	9.25	24.6	28.8
17	4	99.1	−29.8	300	362	20	25	93.7	−54.1	12	4.25	31.7	19.9
18	4	123.8	−43.2	200	1224	65	12	116.0	−53.0	4	3.08	89.2	69.0
19	4	132.1	−53.0	200	871	43	12	131.7	−64.9	11	2.89	36.3	42.2
20*	4	122.3	−46.0	250	108	42	10	119.9	−55.8	3	10.19	66.1	51.0
21	5	100.8	−45.9	300	84	20	20	94.3	−65.5	13	11.97	32.9	33.7
22*	5	97.9	−38.9	200	94	20	20	91.8	−58.2	12	4.93	87.2	42.9
23*	4	122.1	−55.1	300	69	55	10	114.3	−64.0	7	0.76	61.2	26.3
24*	3	122.2	−22.3	400	77	55	20	116.8	−40.4	17	0.047	94.1	88.2
25*	3	130.9	−39.2	150	166	50	20	126.3	−58.7	6	0.037	86.7	72.3
26*	3	127.6	−56.2	300	3634	50	20	112.0	−75.1	18	4.29	63.4	42.2
27	3	122.3	−60.5	400	729	40	10	118.6	−70.3	13	7.90	77.4	72.0
28	3	132.3	−47.2	150	1039	40	7	132.6	−54.1	11	25.80	22.8	49.1
29	3	129.3	−47.5	250	1149	40	7	129.1	−54.5	9	18.00	61.4	67.5
30*	5	98.7	−75.8	300	267	39	6	80.5	−80.4	26	46.74	71.1	36.4

N = Number of samples; D_1, I_1 and D_2, I_2 = Mean declination and inclination before and after structural correction, respectively; δ, ω = strike and dip of volcanics at site; H = Optimum cleaning field in oersted; k = Best precision estimate (Fisher 1953); ∂ = Angular deviation of site direction from mean site direction after structural correction; J_x = Mean sample magnetization intensity (emu/cm^3 × 10^{-3}) after cleaning with a field of x oersted.

TABLE 2. Summary of site mean data for Osler volcanics at optimum cleaning fields,
Location: Nipigon Strait, Lat. 48.7° N, Long. 88.1° W.

SC	N	R	D°	I°	k	α_{95}	φ′	λ′	dp	dm
Lower Reversed Sequence										
No	25	23.5	119.5	−37.8	16.2	7.4				
Yes	25	24.4	115.0	−75.5	40.2	4.6	−165.5°	43.1	5.0	6.8
Upper Normal sequence										
No	5	4.94	299.2	+28.6	68.1	9.3				
Yes	5	4.95	296.5	+39.5	82.3	8.5	+177.9°	33.9	6.1	10.2

N = Number of site means given unit weight; R = Resultant of N unit vectors (Fisher 1953); D° = Declination in degrees east of north; I° = Inclination in degrees, positive downward. k = Best estimate of precision (Fisher 1953); α_{95} = Semiangle of cone of 95% confidence of mean direction; φ′, λ′ = Longitude and latitude of the pole position with the sign convention +φ′ = east, +λ′ = north; dp, dm = Semi-axes of the oval of 95% confidence for the pole position; SC = Structural correction.

in most cases the amount of movement is less than 10° and consistent with the removal of a softer secondary component close to the present axial dipole direction of the earth's field. Compared with the NRM data (Fig. 4A), the scatter in directions progressively decreases after application of optimum magnetic cleaning (Fig. 4B) and subsequent structural corrections (Fig. 4C). Most of the remanence therefore predates folding and is probably primary.

The paleomagnetic results confirm the exist-

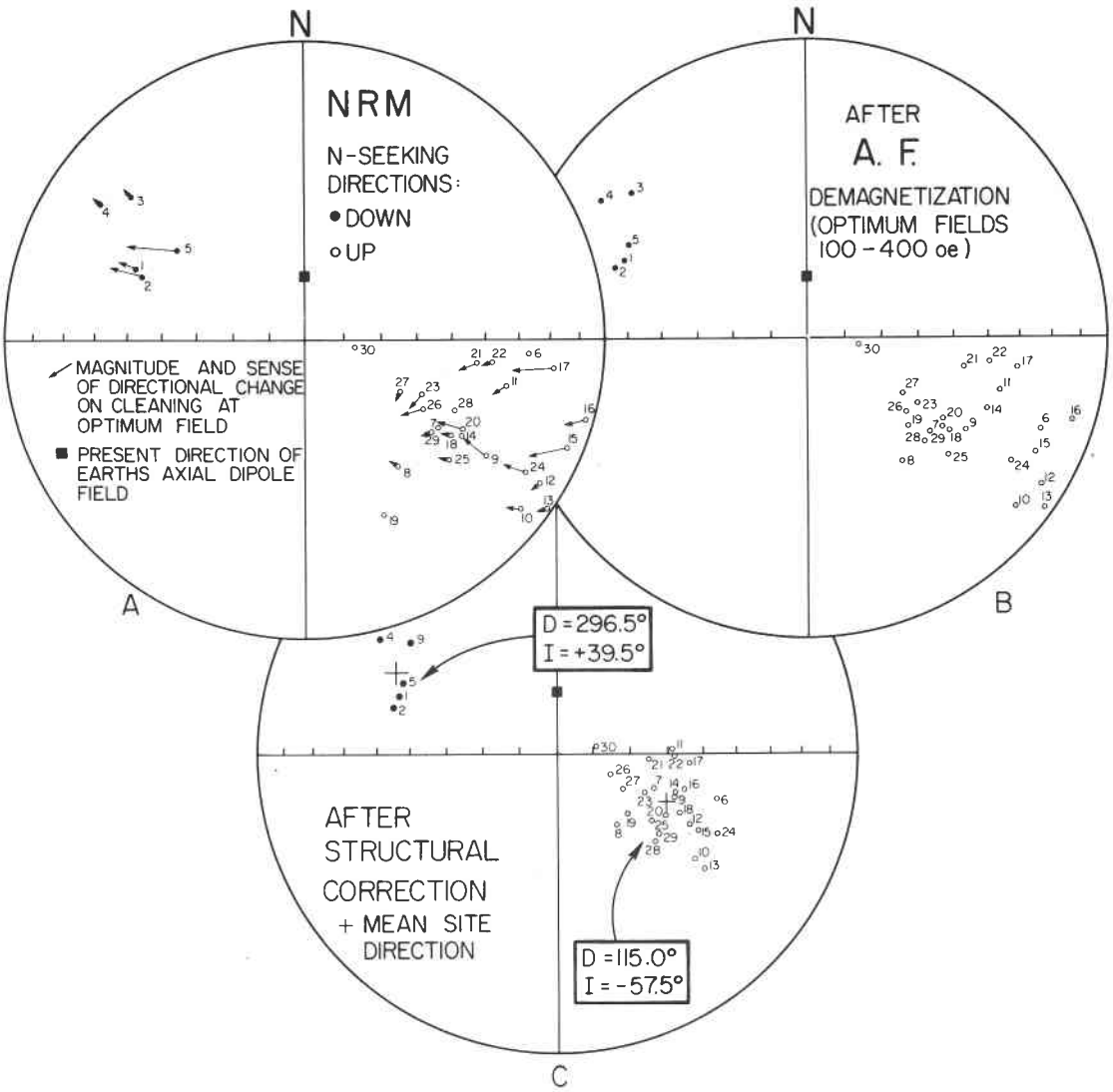


FIG. 4. Equal area projections of site paleomagnetic directions. In diagram A, points without arrows are site directions which either moved erratically or did not change on stepwise cleaning to optimum a.c. magnetic fields.

ence of the reversal postulated (Halls 1972) on aeromagnetic grounds. The position of the reversal coincides exactly with that of the suspected unconformity. Flows above and below the unconformity have normal and reversed magnetization directions respectively. The mean reversed direction yields a pole which is not significantly different from that found by Palmer (1970) in the lower part of the Osler Group, while the corresponding normal pole is similar to that reported from Keweenawan volcanics elsewhere around Lake Superior (*e.g.*, Palmer 1970; Books 1972). The position of the reversal is only about 2 miles (3 km) south from where it was originally drawn on the basis of aeromagnetic data (Halls 1972).

The paleomagnetic results thus amply demonstrate the usefulness of aeromagnetic data in pinpointing an area of interest that would otherwise have remained unsampled due to its inaccessability.

Discussion of Results

Keweenawan volcanic sequences around Lake Superior, together with their associated intrusives have been the subject of numerous paleomagnetic studies (*e.g.*, DuBois 1962; Books 1968, 1972; Palmer 1970; Robertson 1973). The principal result that has emerged from this work is that, with one exception[1], there is no more than a single reversed-to-normal polarity change within any given volcanic sequence around the lake. The reversal is always asymmetric with reversed directions steeper than normal ones. The departure from symmetry ranges from about 15° for volcanics along the northwest shore of the lake to as much as 40° in sequences from the eastern and southern shores. Robertson (1973) has recently summarised evidence which suggests that the reversal asymmetry is caused by polar wander rather than by the addition of a secondary component as advocated by Palmer (1970).

The apparent uniqueness of the reversal in any one volcanic sequence suggests that it is the same one in each case and therefore a valuable time stratigraphic marker. The correlation is further supported by geological observations which show that the position of the reversal also corresponds with a break in volcanism. The new data presented in Fig. 3 offer perhaps the most convincing evidence for an unconformity separating reversed polarity volcanics from younger normal ones. Indications of similar geological relations have also been reported from the Ironwood area in Michigan (Hubbard 1968; White *et al.* 1971). In all but one of the volcanic sequences bordering the lake where the reversal has been observed, including those along the east shore which show no internal discordance, a conglomerate/sandstone unit separates the two volcanic sequences (Hubbard 1967; Palmer 1970). Only in northeast Minnesota is a sedimentary unit absent, but here the Duluth Gabbro Complex has been intruded along the contact and may have destroyed any pre-existing clastic unit.

[1]The exception occurs in the Mamainse Point volcanics (Fig. 1) where both Palmer (1970) and Robertson (1973) have detected *two* reversed-to-normal polarity changes. However both geological and paleomagnetic data favor the hypothesis of a single reversal which has been duplicated by faulting.

Paleomagnetic and geological data thus suggest that throughout the Lake Superior region there was a temporary halt or at least attenuation in volcanic activity. At that time extensive erosion occurred locally around the margins of the subsiding Keweenawan basin and detrital sediments were deposited. Erosion appears to have been more active and/or prolonged in the southern part of the basin, where preserved sediments are thicker. At least 7000 ft (2130 m) accumulated in the Ironwood area (Hubbard 1967) and 2000 ft (610 m) at Mamainse Point, whereas at Cape Gargantua and Nipigon Strait thicknesses are less than 100 ft (30 m).

Conclusions

Geological and paleomagnetic observations from the Nipigon Strait area in northern Lake Superior have verified previous inferences based on aeromagnetic data that an unconformity occurs near the top of the Osler Volcanic Group and separates reversely magnetized lavas from an overlying sequence with normal polarity. The relations found at Nipigon Strait provide further support for the existence of a temporary halt or decline in Keweenawan igneous activity which occurred throughout the Lake Superior region.

Acknowledgments

I would like to thank H. C. Palmer, University of Western Ontario, for permission to use his PAR Spinner Magnetometer during the time that a new bearing assembly was being installed in the one at Toronto. The research was supported by a grant (Number A7824) from the National Research Council of Canada.

Books, K. G. 1968. Magnetization of the lowermost Keweenawan lava flows in the Lake Superior area. U.S. Geol. Surv. Prof. Pap. 600–D, pp. D248–D254.

——— 1972. Paleomagnetism of some Lake Superior Keweenawan rocks. U.S. Geol. Surv. Prof. Pap. 760. 42 p.

Books, K. G. and J. C. Green. 1972. Further paleomagnetic data for Keweenawan rocks in the western Lake Superior area (abstract). Geol. Soc. Am. Abstr. 1972. Ann. Meet., **4**, p. 454.

Chase, C. G. and T. H. Gilmer. 1973. Precambrian plate tectonics: The mid-continent gravity high. Earth Planet. Sci. Lett., **21**, pp. 70–78.

DuBois, P. M. 1962. Paleomagnetism and correlation of Keweenawan rocks. Geol. Surv. Can. Bull. 71, 75 p.

FISHER, R. 1953. Dispersion on a sphere. Proc. Roy. Soc. London, Series A, **217**, pp. 295–305.

HALLS, H. C. 1966. A review of the Keweenawan geology of the Lake Superior region. *In*: The earth beneath the continents. Am. Geophys. Un. Mono. 10, pp. 3–27.

——— 1972. Magnetic studies in northern Lake Superior. Can. J. Earth Sci., **9**, pp. 1349–1367.

HALLS, H. C. and G. F. WEST. 1971. A seismic refraction survey in Lake Superior. Can. J. Earth Sci., **8**, pp. 610–630.

HUBBARD, H. A. 1967. Keweenawan volcanic rocks near Ironwood, Michigan. (abstr.). 13th Ann. Inst. Lake Superior Geology, Michigan State U., East Lansing, p. 20.

——— 1968. Stratigraphic relationships of some Keweenawan rocks of Michigan and Wisconsin (abstr.). 14th Ann. Inst. Lake Superior Geology, Wisconsin State U., Superior, pp. 35–36.

MCILWAINE, W. H., WALLACE, H., and D'APOLLONIA, S. J. 1971. Black Bay Peninsula and vicinity. Ont. Dept. Mines, Prelim. Geol. Map Nos. p. 622 (with Kustra C. R.); P. 623; P. 624 (with Keeler R. G.); P. 625–28 Inc.

PALMER, H. C. 1970. Paleomagnetism and correlation of some Middle Keweenawan rocks, Lake Superior. Can. J. Earth Sci., **7**, pp. 1410–1436.

ROBERTSON, W. A. 1973. Pole positions from the Mamainse Point lavas and their bearing on a Keweenawan pole path and polarity sequence. Can. J Earth Sci., **10**, p. 1541–1555.

SMITH, T. J., STEINHART, J. S., and ALDRICH, L. T. 1966. Crustal structure under Lake Superior. *In*: The Earth beneath the continents. Am. Geophys. Un. Mono. 10, pp. 181–197.

WHITE, W. S. 1966. Tectonics of the Keweenawan basin, western Lake Superior region. U.S. Geol. Surv. Prof. Pap. 524-E, 23 p.

——— 1972. Keweenawan flood basalts and continental rifting (abstr.). Geol. Soc. Am. Abstr. 1972 Ann. Meet., 4, pp. 732–734.

WHITE, W. S., BROOKS, E. R., HUBBARD, H. A., JOHNSON, R. F. and WILBAND, J. T. 1971. Keweenawan stratigraphy of westernmost Michigan (abstr.). 17th Ann Inst. Lake Superior Geology. pp. 71–72.